Drivers Behind New Network Architectures

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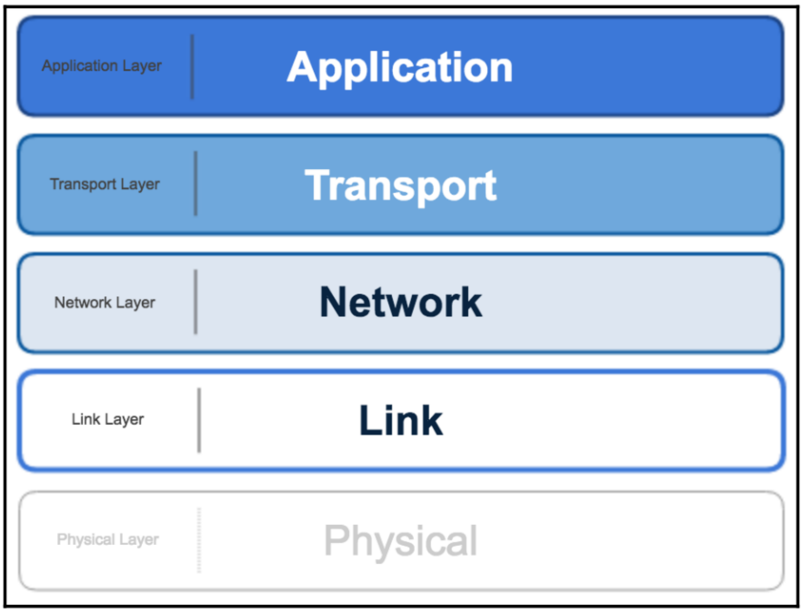
1 Networking basics

Here is an oversimplified view of the networking protocol stack. There is much more going on here than we will cover in this chapter, but to help in understanding the discussion, it is useful to reduce it to a simple diagram:



Network communications operates in layers with the bottom layers not needing to know about the layers above it. It can get confusing talking about all the options available at each layer of the stack. The diagram shows the key layers that we are concerned with for IoT analytics, but know that there is more to the story.

The diagram is based on the simplified OSI model, which divides communication into five fundamental layers. There is a Physical layer at bottom that has more to do with device electrical engineering. We will leave that out to simplify things since we are focusing on analytics.

Connectivity will refer to options primarily in the Link layer of the stack. Data communication or messaging will be referring mainly to the Application layer. The Network layer is (usually) the Internet Protocol (IP) that we all know and love. It is a great consolidator that enables so much flexibility in the networking stack for internet communications.

Transport is either TCP or UDP for most networking schemes that transmit data over long distances. TCP has more delivery guarantees but more overhead; UDP has less overhead but less guarantees on delivery.

There is a wide variety of networking protocols for both connectivity from the internet to the IoT device and for the communication of the data the device generates. They solve two main problems:

- **How do I establish communication, so I can send data packets?**
    - Let's use snail mail as an analogy. If you have limited funds, you prefer sending brief postcards even it takes longer to arrive to the recipient. It costs less, and it does not take much effort to write. If money is not a problem, you would hire some professional authors and send prose approaching poetry. You would use airmail so it arrives quickly. In this analogy, battery power is like money and CPU/memory is like writing effort.

- **How are the data packets delivered so that my needs are met?**
    - Do you and I agree to get each other's address and send letters direct? Or do you always give them to Tom, in the cubicle down the hall, and he sends it out to whoever needs it? Do you send it certified mail because you have to be certain it was delivered? Or do you send it third-class post because it is much cheaper and not the end of the world if it gets lost?

The network protocol used depends on the type of device and the environment in which it is located. It can also depend on the business priorities for which the data will be used.

For example, in the case of monitoring aircraft engines in flight, it is more important to get the most recent data than to get all of the data. You need problems identified as soon as possible more than you need a complete recording of all in flight data, which is mostly the same. So if you have to make a choice between waiting but getting everything or never waiting but losing some data packets from time to time, you choose the latter.

This is different from the logistics provider that has a sensor, which detects tractor trailers entering and leaving a distribution hub. In this case, you want a complete record of traffic, and you would be willing to wait a minute or two to have it.

Another important component to understand for IoT is that of a **gateway**. A gateway is a network node that connects two separate networks that use different protocols. The gateway handles the translation between them. A gateway is common in many IoT networks of constrained devices. It translates from the protocol optimized for the constrained devices to (usually) the protocol used by the wider internet (IP).

There are multiple standards spread across several organizations such as the Industrial Internet Consortium, the IPSO Alliance, and the IEEE P2413. It is continuously evolving with many companies and alliances aligned along different connectivity protocols. It may be another five to 10 ten years before the options coalesce into a smaller group of consistently used standards.

2 Radio Frequency IDentification (RFID)

RFID system is able to overcome most of these difficulties to an extent. The RFID system consists of a tag (also known as a transponder) attached to the object being identified. The tag usually consists of an integrated circuit and an antenna. Another important module in the system is a reader. The reader queries the tag using radio frequency (RF) waves, and gets the identity of the tag via the RF waves. The RFID systems operate in various frequency bands. Some of the most popular frequencies are:

- 125 kHz to 134.2 kHz (LF: low frequency);
- 13.56 MHz (HF: high frequency);
- 860 to 915 MHz (UHF: ultra-high frequency); and
- 2.45 GHz to 5.8 GHz (microwave frequency).

The RFID systems operating in the LF band were the first to be deployed in the market for high-volume short-range industrial applications and car immobilizer devices. These systems are attractive in systems where the data rates are not very high. The HF RFID systems are capable of handling much higher data rates compared to the LF system, and the tag antenna is much smaller. HF systems have longer read range compared to the LF systems. The UHF RFID system has a much longer read range and much higher data rate compared to the LF and HF systems. However, the UHF system does not work very well in the presence of metallic objects, water and the human body, compared to the LF system.

2.1 Principle of RFID

Consider a coil made of copper wire through which alternating current is flowing. The coil offers impedance to the source and a voltage develops across its terminals. It is possible to increase the voltage by connecting a capacitor in parallel with the coil. Let us call this the "primary" coil. Now we bring in another coil, called the "secondary" coil, close to the first. Due to electromagnetic induction, voltage appears across the terminals of the secondary coil. The amplitude of the voltage depends on the size, shape, location and orientation of the secondary coil. If we connect a resistor (also known as a load) across the terminals of the secondary coil, current flows through it. The strength of the current flowing through the secondary coil depends on the load. The interesting phenomenon is that the current flowing in the secondary coil induces a voltage back into the primary coil, which is proportional to its strength. The induced voltage, also known as back emf (electromotive force), can easily be sensed by using suitable electronics. Therefore, by observing the voltage on the primary, it is possible to estimate what is connected to the secondary coil.

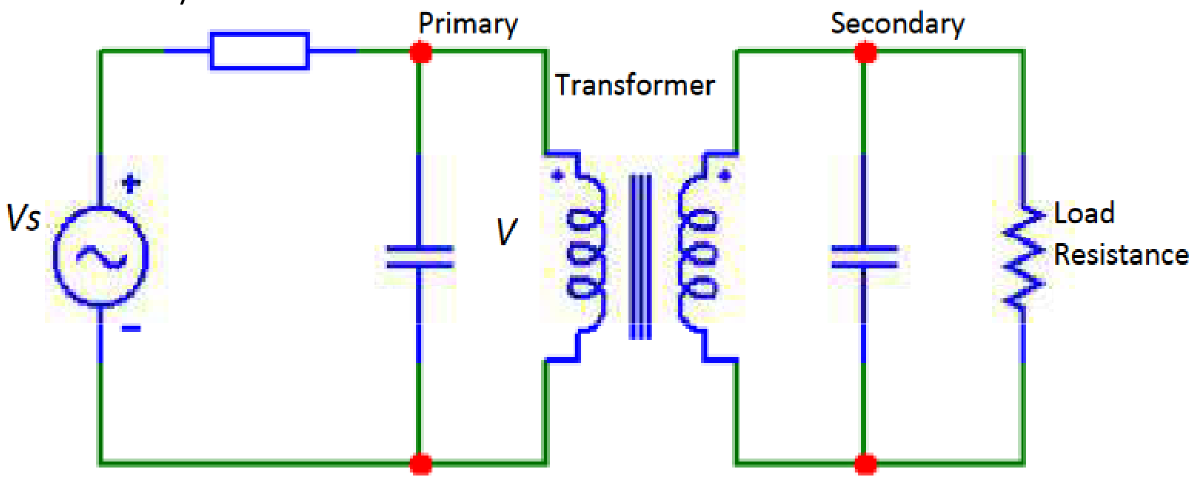


Figure 1: Circuit schematic of two coils electromagnetically interacting with each other

3 NFC (Near-Field Communication)

NFC allows two devices to communicate when brought within very short distances of each other - about 4 cm. It is a set of communication protocols that offer low speeds and simple setup. There is a variety of NFC protocol stacks depending on the type and use case.

Figure 2: NFC Logo

NFC uses electromagnetic induction between two loop style antennas when devices exchange information. Data rates are around 100-400 kbit/s. It operates over radio frequency band 13.56 MHz, which is unlicensed and globally available.

There is usually a full NFC device, such as a smartphone, and an NFC tag, such as a chip reader. All full NFC devices can operate in three modes:

- **NFC peer-to-peer:** Devices exchange information on the fly.
- **NFC card emulation:** Smartphones can act like smart cards in this mode, allowing mobile payments or ticketing.
- **NFC reader/writer:** Full NFC devices can read information stored in inexpensive NFC tags, such as the type used on posters or pamphlets. You have probably used this at a conference to scan your attendee badge.

NFC tags are usually read-only, but it is possible for them to be writeable. There is an initiator device and a target device in NFC communications. The initiator device creates an RF field that can power a target device. This means NFC tags do not need batteries in what is called passive mode. Unpowered NFC tag devices are very inexpensive, just a few pennies.

There is also an active mode where each device can communicate with the other by alternatively generating its own field to read data, then turning it off to receive data. This mode typically requires both the devices to have a power supply.

For data records that eventually end up in an analytics dataset, the full NFC device would act as a gateway, translate the message, and then communicate over a different network protocol stack. In the case of a smartphone, it would take the NFC tag data and create its own data message to send over 4G/LTE for a destination on the internet.

3.1 Common use cases

NFC is commonly used in the following applications:

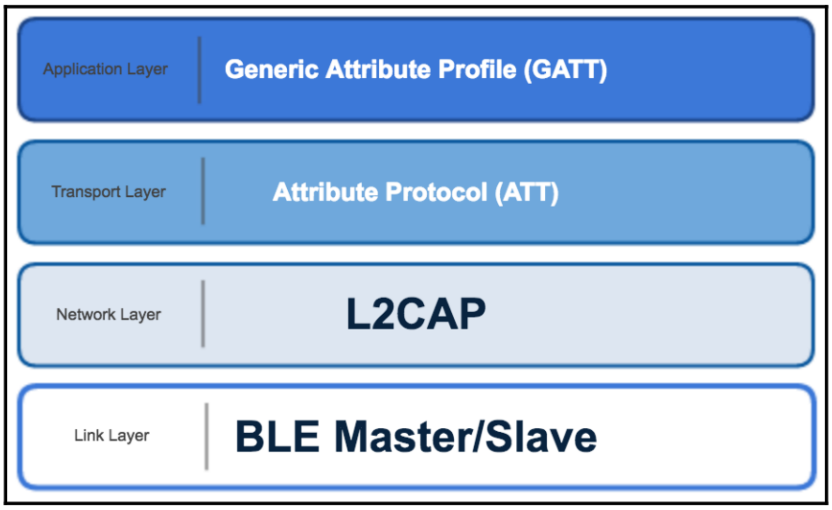
- Smartphone payments:
  - Apple pay
  - Host Card Emulation (HCE) - Android supported
- Smart posters
- Identification cards
- Event logging
- Triggering a programmable action

4 Bluetooth Low Energy (BLE) roles

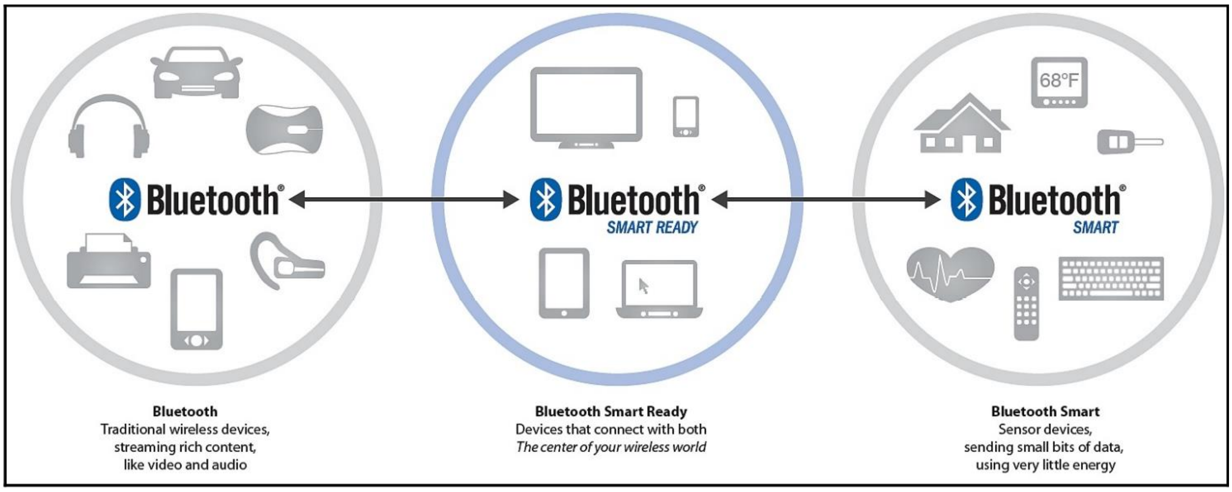
Named after a tenth-century Danish king who unified several fractious tribes, Bluetooth is a familiar technology to most consumers. The specification is maintained by an organization called the Bluetooth Special Interest Group (SIG) made up of 25,000 companies. Bluetooth Low Energy (BLE) is a newer implementation that is not directly compatible with Bluetooth classic (of mobile phone headset fame). It was designed for low power needs and a less frequent data exchange rate. The following chart represents the network protocol stack for BLE:

A BLE device is either a master or a slave in the network. A master acts as an advertiser in the connection while the slave acts as receiver. A master can have several slaves connected to it while a slave can link to only one master. The network group involving the slaves connected to a master is called a **piconet**.

To make data available for analytics, the master device passes it over the internet using one of the data messaging protocols that will be discussed later in this chapter. There is typically a gateway device that is both part of the bluetooth network and also linked to the internet. It translates data reported from BLE nodes into the appropriate format for data messaging over the internet. It is commonly an unconstrained device that does not need to worry much about power.



Bluetooth 5.0 specification released mid-2016 promises to double the speed, quadruple the range, and bring an eight-fold increase in the capability for data broadcasting for low-power scenarios.



5 LiFi (light fidelity)

Li-Fi (also written as LiFi) is a wireless communication technology which utilizes light to transmit data and position between devices. The term was first introduced by Harald Haas during a 2011 TEDGlobal talk in Edinburgh.

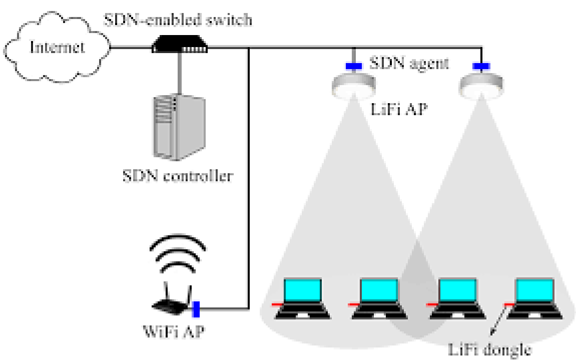


Figure 3: Li-Fi network

Li-Fi is a light communication system that is capable of transmitting data at high speeds over the visible light, ultraviolet, and infrared spectrums. In its present state, only LED lamps can be used for the transmission of data in visible light.

In terms of its end user, the technology is similar to Wi-Fi — the key technical difference being that Wi-Fi uses radio frequency to induce a voltage in an antenna to transmit data, whereas Li-Fi uses the modulation of light intensity to transmit data. Li-Fi is able to function in areas otherwise susceptible to electromagnetic interference (e.g., aircraft cabins, hospitals, or the military).

Bg-Fi is a Li-Fi system consisting of an application for a mobile device, and a simple consumer product, like an IoT (Internet of Things) device, with color sensor, microcontroller, and embedded software. Light from the mobile device display communicates to the color sensor on the consumer product, which converts the light into digital information. Light-emitting diodes enable the consumer product to communicate synchronously with the mobile device.

6 IEEE 802.15 (WPAN standards)

IEEE 802.15 is a working group of the Institute of Electrical and Electronics Engineers (IEEE) IEEE 802 standards committee which specifies wireless personal area network (WPAN) standards. There are 10 major areas of development, not all of which are active.

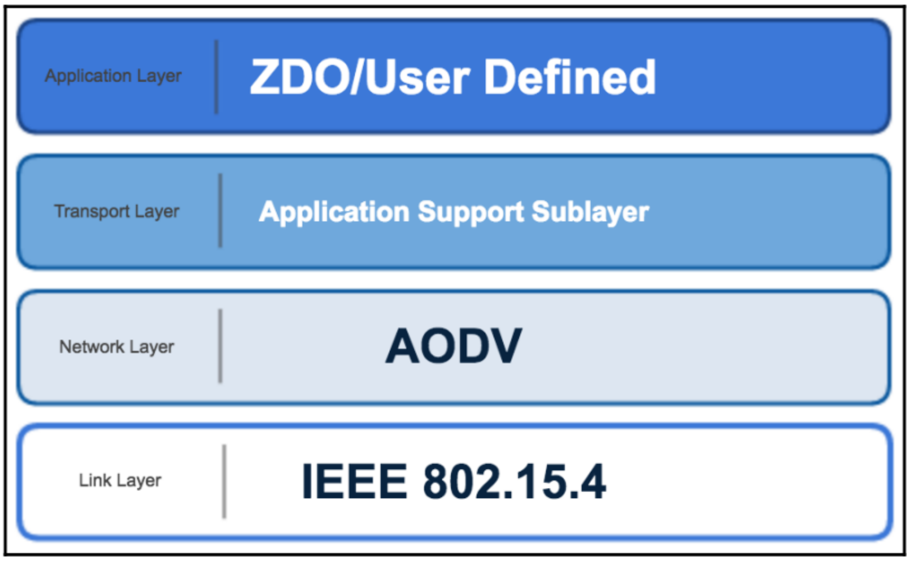
The number of Task Groups in IEEE 802.15 varies based on the number of active projects. The current list of active projects can be found on the IEEE 802.15 web site.

IEEE 802.15.1 (Bluetooth)

7 IEEE 802.15.4 (Low Rate WPAN)

7.1 Zigbee

ZigBee is a little different animal than what we have talked about so far. It covers multiple layers of the protocol stack. It is the network protocol, the transport layer, and an application framework that is the underpinnings of a customer developed application layer.



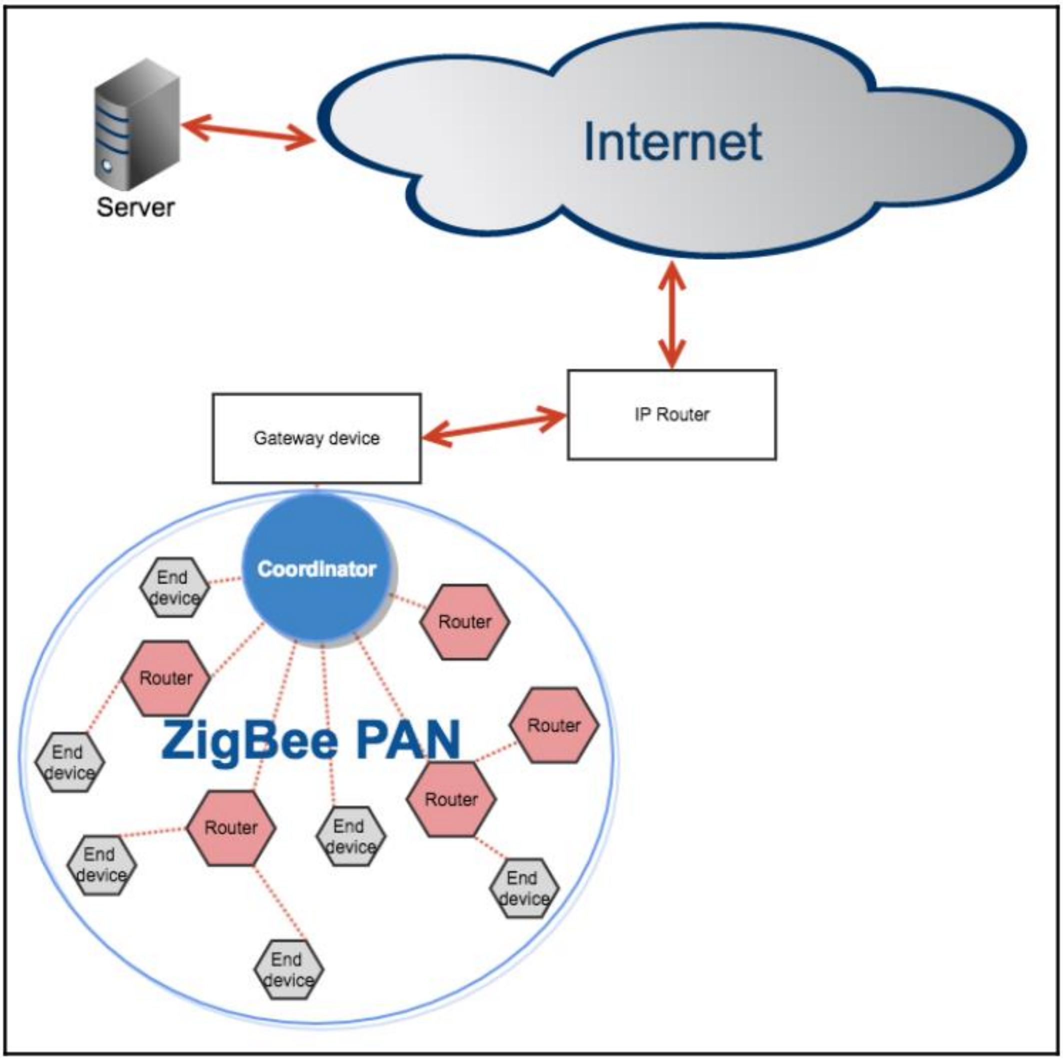
It has security built-in as a requirement and works using small, low power digital radios. It is a self-organizing wireless mesh-style network. The name comes from the waggle dance honey bees often do on returning to the hive.

The range is short, up to 100 meters line of sight, but it can travel long distances by passing data through a mesh network of other ZigBee devices.

The ZigBee alliance is a consortium of companies that maintain and publish Zigbee standards. ZigBee was first released as a standard in 2003. It is intended to be simpler and cheaper than other wireless protocols. Development code can be as little as 2% of equivalent code on Bluetooth.

Data rates are low and devices tend to be simple. ZigBee standards require at least a two-year battery life for a device to qualify. The data transmission is periodic and intermittent. There are different categories of ZigBee device. A fully functioning device can act as a network coordinator and operates in both star and mesh topologies. A reduced functionality device can only communicate with a network coordinator and works in a star topology.

These devices form a **Personal Area Network (PAN).** Network interaction times can be very fast. A device can join a network in 30 ms and change from sleeping to active in 15 ms.



Example of ZigBee network

There are three types of ZigBee devices in a network:

A **coordinator device** that initiates network formation. There is one and only of these for each network. It may also act as a router once the network is formed. It serves as the trust center and is the repository for security keys.

A **router device** is optional. It can join with a coordinator or another router device and acts as a multi-hop intermediary in routing of messages.

An **end device** has just enough functionality to communicate with its coordinator. It is optional and cannot participate in the routing of messages.

The ZigBee networking routing protocol is reactive. It establishes a route path on demand instead of proactively such as IP. The network protocol used is **Ad hoc On-Demand Distant Vector (AODV)** which finds a destination by broadcasting to all its neighbours, which then send a request to all their neighbours, until the destination is found.

The network is silent between connections. Networks are self-healing. ZigBee networks are secured by 128-bit AES encryption. There is often a gateway device that is part of the ZigBee network and also linked to the internet. It translates data into the appropriate format for messaging over the internet. It can also convert ZigBee network routing protocol into IP routing protocol. The gateway is commonly an unconstrained device that does not need to worry much about power.

7.1.1 Advantages of ZigBee

There are several advantages to ZigBee networks:

- **Reliable and robust:** Networks are self-forming and self-healing.
- **Low power requirements:** Devices can operate for years without needing a new battery. Solar power could last even longer.
- **Global:** The wireless protocol it is built on is available globally.
- **Security:** Encryption is part of the standard instead of an optional add on.

7.1.2 Disadvantages of ZigBee

There are also some disadvantages:

- **Not free:** It costs $3,500 USD at the time of writing to license the standard
- **Low data rates:** It is not designed to support higher data rates
- **Star network is limited:** The coordinator device supports up to 65,000 devices

7.1.3 Common use cases

ZigBee is commonly used in the following applications:

- Home automation
- Utilities monitoring and control
- Smart lighting
- Building automation
- Security systems
- Medical data collection

7.2 Z-wave

Z-Wave is a wireless communications protocol used primarily for residential and commercial building automation. It is a mesh network using low-energy radio waves to communicate from device to device, allowing for wireless control of smart home devices, such as smart lights, security systems, thermostats, sensors, smart door locks, and garage door openers. Like other protocols and systems aimed at the residential, commercial, MDU and building markets, a Z-Wave system can be controlled from a smart phone, tablet, or computer, and locally through a smart speaker, wireless keyfob, or wall-mounted panel with a Z-Wave gateway or central control device serving as both the hub or controller. Z-Wave provides the application layer interoperability between home control systems of different manufacturers that are a part of its alliance. There is a growing number of interoperable Z-Wave products; over 1,700 in 2017, over 2,600 by 2019, and over 4,000 by 2022.

8 Narrow Band IoT

Recognizing that the definition of new LTE device categories was not sufficient to support LPWA IoT requirement, 3GPP specified Narrowband IoT (NB-IoT). The work on NB-IoT started with multiple proposals pushed by the involved vendors, including the following:

- Extended Coverage GSM (EC-GSM), Ericsson proposal
- Narrowband GSM (N-GSM), Nokia proposal
- Narrowband M2M (NB-M2M), Huawei/Neul proposal
- Narrowband OFDMA (orthogonal frequency-division multiple access), Qualcomm proposal
- Narrowband Cellular IoT (NB-CIoT), combined proposal of NB-M2M and NB-OFDMA
- Narrowband LTE (NB-LTE), Alcatel-Lucent, Ericsson, and Nokia proposal
- Cooperative Ultra Narrowband (C-UNB), Sigfox proposal

Consolidation occurred with the agreement to specify a single NB-IoT version based on orthogonal frequency-division multiple access (OFDMA) in the downlink and a couple options for the uplink. OFDMA is a modulation scheme in which individual users are assigned subsets of subcarrier frequencies. This enables multiple users to transmit lowspeed data simultaneously. For more information on the uplink options, refer to the 3GPP specification TR 36.802.

Three modes of operation are applicable to NB-IoT:

- **Standalone:** A GSM carrier is used as an NB-IoT carrier, enabling reuse of 900 MHz or 1800 MHz.
    - **In-band:** Part of an LTE carrier frequency band is allocated for use as an NB-IoT frequency. The service provider typically makes this allocation, and IoT devices are configured accordingly. You should be aware that if these devices must be deployed across different countries or regions using a different service provider, problems may occur unless there is some coordination between the service providers, and the NB-IoT frequency band allocations are the same.
    - **Guard band:** An NB-IoT carrier is between the LTE or WCDMA bands. This requires coexistence between LTE and NB-IoT bands.

In its Release 13, 3GPP completed the standardization of NB-IoT. Beyond the radio specific aspects, this work specifies the adaptation of the IoT core to support specific IoT capabilities, including simplifying the LTE attach procedure so that a dedicated bearer channel is not required and transporting non-IP data. Subsequent releases of 3GPP NB-IoT will introduce additional features and functionality, such as multicasting, and will be backward compatible with Release 13.

Mobile service providers consider NB-IoT the target technology as it allows them to leverage their licensed spectrum to support LPWA use cases. For instance, NB-IoT is defined for a 200-kHz-wide channel in both uplink and downlink, allowing mobile service providers to optimize their spectrum, with a number of deployment options for GSM, WCDMA, and LTE spectrum, as shown in Figure 4-19.

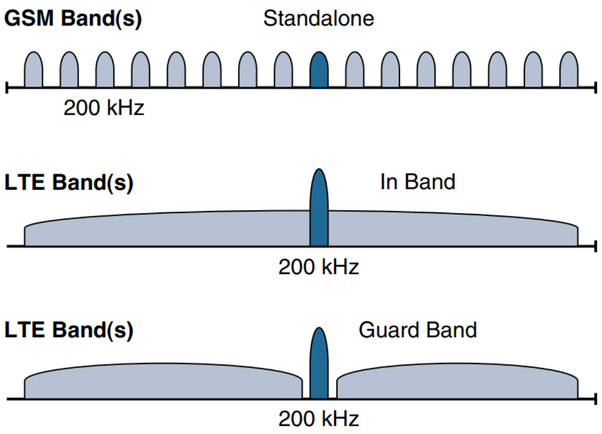


Figure 4-19 *NB-IoT Deployment Options*

In an LTE network, resource blocks are defined with an effective bandwidth of 180 kHz, while on NB-IoT, tone or subcarriers replace the LTE resource blocks. The uplink channel can be 15 kHz or 3.75 kHz or multi-tone (n*15 kHz, n up to 12). At Layer 1, the maximum transport block size (TBS) for downlink is 680 bits, while uplink is 1000 bits. At Layer 2, the maximum Packet Data Convergence Protocol (PDCP) service data unit (SDU) size is 1600 bytes.

NB-IoT operates in half-duplex frequency-division duplexing (FDD) mode with a maximum data rate uplink of 60 kbps and downlink of 30 kbps.

8.1 Topology

NB-IoT is defined with a link budget of 164 dB; compare this with the GPRS link budget of 144 dB, used by many machine-to-machine services. The additional 20 dB link budget increase should guarantee better signal penetration in buildings and basements while achieving battery life requirements.

8.2 Competitive Technologies

In licensed bands, it is expected that 3GPP NB-IoT will be the adopted LPWA technology when it is fully available. Competitive technologies are mostly the unlicensed-band LPWA technologies such as LoRaWAN. The main challenge faced by providers of the licensed bands is the opportunity for non-mobile service providers to grab market share by offering IoT infrastructure without buying expensive spectrum

Internet Protocol and Transmission Control Protocol

6LoWPAN

WLAN and WAN

IEEE 802.11

Long-range Communication Systems and Protocols:

9 Cellular Connectivity-LTE

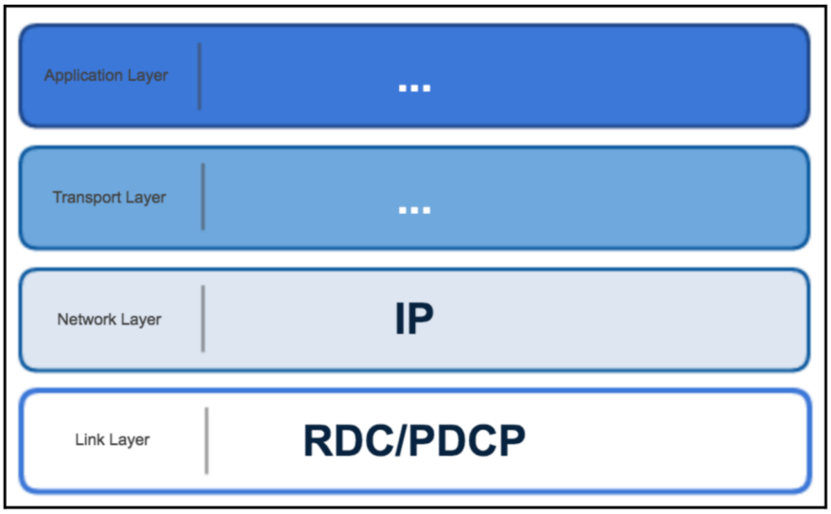
Cellular (4G/LTE) data operates over radio waves. The fourth-generation networking technology, 4G, is an IP-based networking architecture handling both voice and data. Long Term Evolution (LTE) is a 4G technology. It handles data rates up to 300 Mbit/s.

4G LTE logo

LTE can handle fast moving mobile devices (such as mobile phones or telematics units). It can handle devices moving out of range of one network and into another without disruption. Due to higher power needs, it usually requires either constant power or frequent battery charges. LTE supports different categories of services.

A newer category, **LTE Category 0 (LTE Cat 0)**, is designed for IoT requirements. It has lower data rates, capped at 1 Mbit/s maximum. LTE Cat 0 also reduces bandwidth ranges, which allows for significant complexity reduction. It is still able to operate on all existing LTE networks and supports low power constrained IoT devices.

Cellular coverage is not available in all areas, though. There are still some places where you will not find a signal. For analytics, this means that if your IoT device moves around, it could venture into an area without a signal. Data could be lost in this case, unless it is buffered until the signal is recovered.



9.1 Common use cases

Cellular is commonly used in the follow applications:

- Freight tracking
- Vehicle connectivity (telematics)
- Smartphones (obviously)
- Remote equipment monitoring.

9.2 LTE Cat 0

The first enhancements to better support IoT devices in 3GPP occurred in LTE Release 12. A new user equipment (UE) category, Category 0, was added, with devices running at a maximum data rate of 1 Mbps. Generally, LTE enhancements target higher bandwidth improvements. Category 0 includes important characteristics to be supported by both the network and end devices. Meanwhile, the UE still can operate in existing LTE systems with bandwidths up to 20 MHz. These Cat 0 characteristics include the following:

- **Power saving mode (PSM):** This new device status minimizes energy consumption. Energy consumption is expected to be lower with PSM than with existing idle mode. PSM is defined as being similar to "powered off" mode, but the device stays registered with the network. By staying registered, the device avoids having to re attach or re-establish its network connection. The device negotiates with the network the idle time after which it will wake up. When it wakes up, it initiates a tracking area update (TAU), after which it stays available for a configured time and then switches back to sleep mode or PSM. A TAU is a procedure that an LTE device uses to let the network know its current tracking area, or the group of towers in the network from which it can be reached. Basically, with PSM, a device can be practically powered off but not lose its place in the network.
- **Half-duplex mode:** This mode reduces the cost and complexity of a device's implementation because a duplex filter is not needed. Most IoT endpoints are sensors that send low amounts of data that do not have a full-duplex communication requirement.

9.3 LTE-M

Following LTE Cat 0, the next step in making the licensed spectrum more supportive of IoT devices was the introduction of the LTE-M category for 3GPP LTE Release 13. These are the main characteristics of the LTE-M category in Release 13:

Lower receiver bandwidth: Bandwidth has been lowered to 1.4 MHz versus the usual 20 MHz. This further simplifies the LTE endpoint.

Lower data rate: Data is around 200 kbps for LTE-M, compared to 1 Mbps for Cat 0.

Half-duplex mode: Just as with Cat 0, LTE-M offers a half-duplex mode that decreases node complexity and cost.

Enhanced discontinuous reception (eDRX): This capability increases from seconds to minutes the amount of time an endpoint can "sleep" between paging cycles. A paging cycle is a periodic check-in with the network. This extended "sleep" time between paging cycles extends the battery lifetime for an endpoint significantly.

10 LoRaWAN

In recent years, a new set of wireless technologies known as Low-Power Wide-Area (LPWA) has received a lot of attention from the industry and press. Particularly well adapted for long-range and battery-powered endpoints, LPWA technologies open new business opportunities to both services providers and enterprises considering IoT solutions. This section discusses an example of an unlicensed-band LPWA technology, known as LoRaWAN, and the next section, "NB-IoT and Other LTE Variations," reviews licensed-band alternatives from the 3rd Generation Partnership Project (3GPP).

10.1 Standardization and Alliances

Initially, LoRa was a physical layer, or Layer 1, modulation that was developed by a French company named Cycleo. Later, Cycleo was acquired by Semtech. Optimized for long-range, two-way communications and low power consumption, the technology evolved from Layer 1 to a broader scope through the creation of the LoRa Alliance. For more information on the LoRa Alliance, visit www.lora-alliance.org.

The LoRa Alliance quickly achieved industry support and currently has hundreds of members. Published LoRaWAN specifications are open and can be accessed from the LoRa Alliance website.

Semtech LoRa as a Layer 1 PHY modulation technology is available through multiple chipset vendors. To differentiate from the physical layer modulation known as LoRa, the LoRa Alliance uses the term LoRaWAN to refer to its architecture and its specifications that describe end-to-end LoRaWAN communications and protocols.

Figure 4-15 provides a high-level overview of the LoRaWAN layers. In this figure, notice that Semtech is responsible for the PHY layer, while the LoRa Alliance handles the MAC layer and regional frequency bands.

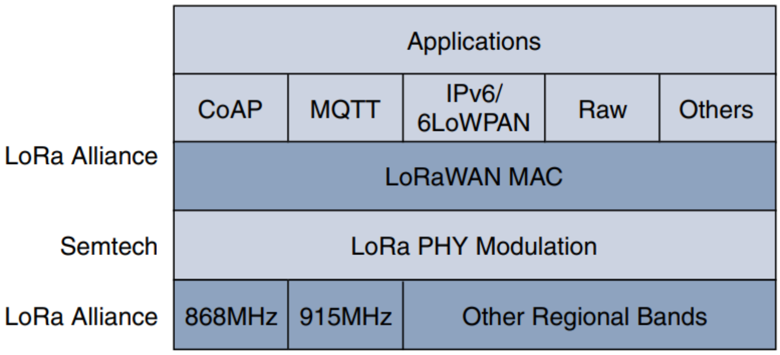


Figure 4-15 *LoRaWAN Layers*

Overall, the LoRa Alliance owns and manages the roadmap and technical development of the LoRaWAN architecture and protocol. This alliance also handles the LoRaWAN endpoint certification program and technology promotion through its certification and marketing committees.

10.2 Physical Layer

Semtech LoRa modulation is based on chirp spread spectrum modulation, which trades a lower data rate for receiver sensitivity to significantly increase the communication distance. In addition, it allows demodulation below the noise floor, offers robustness to noise and interference, and manages a single channel occupation by different spreading factors. This enables LoRa devices to receive on multiple channels in parallel.

LoRaWAN 1.0.2 regional specifications describe the use of the main unlicensed sub-GHz frequency bands of 433 MHz, 779–787 MHz, 863–870 MHz, and 902–928 MHz, as well as regional profiles for a subset of the 902–928 MHz bandwidth. For example, Australia utilizes 915–928 MHz frequency bands, while South Korea uses 920–923 MHz and Japan use 920–928 MHz.

Understanding LoRa gateways is critical to understanding a LoRaWAN system. A LoRa gateway is deployed as the center hub of a star network architecture. It uses multiple transceivers and channels and can demodulate multiple channels at once or even demodulate multiple signals on the same channel simultaneously. LoRa gateways serve as a transparent bridge relaying data between endpoints, and the endpoints use a single-hop wireless connection to communicate with one or many gateways.

Table 4-4 *LoRaWAN Data Rate Example*

Configuration	863–870 MHz bps	902–928 MHz bps
LoRa: SF12/125 kHz	250	N/A
LoRa: SF11/125 kHz	440	N/A
LoRa: SF10/125 kHz	980	980
LoRa: SF9/125 kHz	1760	1760
LoRa: SF8/125 kHz	3125	3125
LoRa: SF7/125 kHz	5470	5470
LoRa: SF7/250 kHz	11,000	N/A
FSK: 50 kbps	50,000	N/A
LoRa: SF12/500 kHz	N/A	980
LoRa: SF11/500 kHz	N/A	1760
LoRa: SF10/500 kHz	N/A	3900
LoRa: SF9/500 kHz	N/A	7000
LoRa: SF8/500 kHz	N/A	12,500
LoRa: SF7/500 kHz	N/A	21,900

The data rate in LoRaWAN varies depending on the frequency bands and adaptive data rate (ADR). ADR is an algorithm that manages the data rate and radio signal for each endpoint. The ADR algorithm ensures that packets are delivered at the best data rate possible and that network performance is both optimal and scalable. Endpoints close to the gateways with good signal values transmit with the highest data rate, which enables a shorter transmission time over the wireless network, and the lowest transmit power. Meanwhile, endpoints at the edge of the link budget communicate at the lowest data rate and highest transmit power.

An important feature of LoRa is its ability to handle various data rates via the spreading factor. Devices with a low spreading factor (SF) achieve less distance in their communications but transmit at faster speeds, resulting in less airtime. A higher SF provides slower transmission rates but achieves a higher reliability at longer distances. Table 4-4 illustrates how LoRaWAN data rates can vary depending on the associated spreading factor for the two main frequency bands, 863–870 MHz and 902–928 MHz.

In Table 4-4, notice the relationship between SF and data rate. For example, at an SF value of 12 for 125 kHz of channel bandwidth, the data rate is 250 bps. However, when the SF is decreased to a value of 7, the data rate increases to 5470 bps.

Channel bandwidth values of 125 kHz, 250 kHz, and 500 kHz are also evident in Table 4-4. The effect of increasing the bandwidth is that faster data rates can be achieved for the same spreading factor.

10.3 MAC Layer

As mentioned previously, the MAC layer is defined in the LoRaWAN specification. This layer takes advantage of the LoRa physical layer and classifies LoRaWAN endpoints to optimize their battery life and ensure downstream communications to the LoRaWAN endpoints. The LoRaWAN specification documents three classes of LoRaWAN devices:

- **Class A:** This class is the default implementation. Optimized for battery-powered nodes, it allows bidirectional communications, where a given node is able to receive downstream traffic after transmitting. Two receive windows are available after each transmission.
- **Class B:** This class was designated "experimental" in LoRaWAN 1.0.1 until it can be better defined. A Class B node or endpoint should get additional receive windows compared to Class A, but gateways must be synchronized through a beaconing process.
- **Class C:** This class is particularly adapted for powered nodes. This classification enables a node to be continuously listening by keeping its receive window open when not transmitting.

10.4 Topology

LoRaWAN topology is often described as a "star of stars" topology. As shown in Figure 4-17, the infrastructure consists of endpoints exchanging packets through gateways acting as bridges, with a central LoRaWAN network server. Gateways connect to the backend network using standard IP connections, and endpoints communicate directly with one or more gateways.

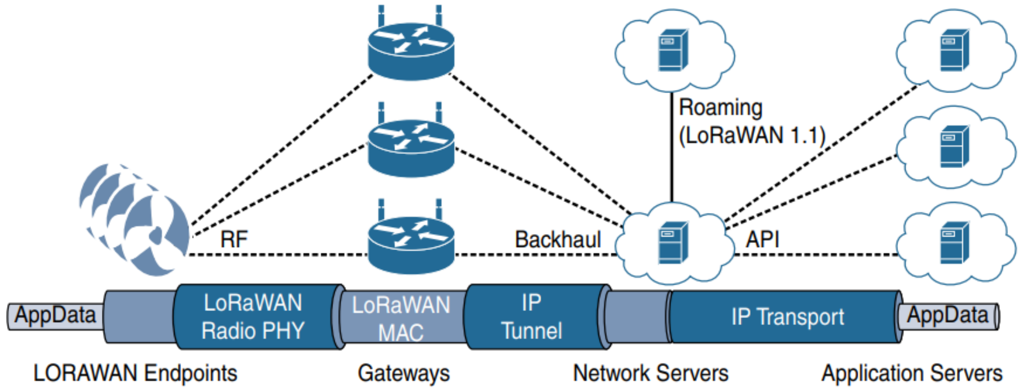


Figure 4-17 *LoRaWAN Architecture*